

BLDC MOTOR

A PROTOTYPE AND ANALYSIS



IIT BHUBANESWAR



TEAM COMPOSITION

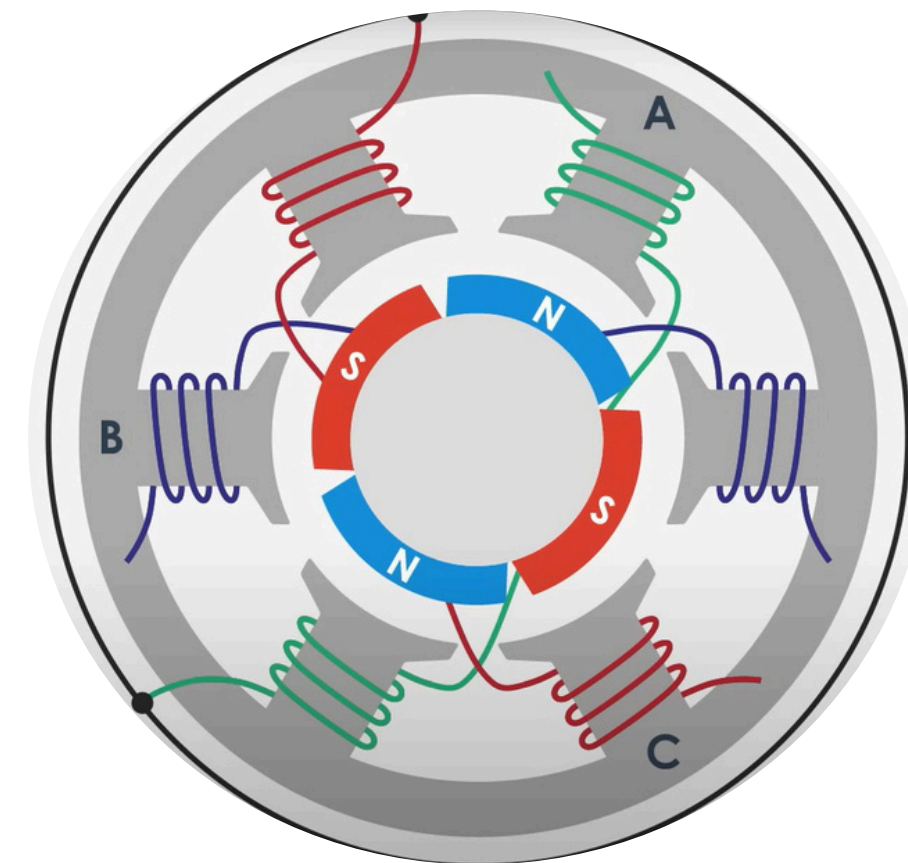
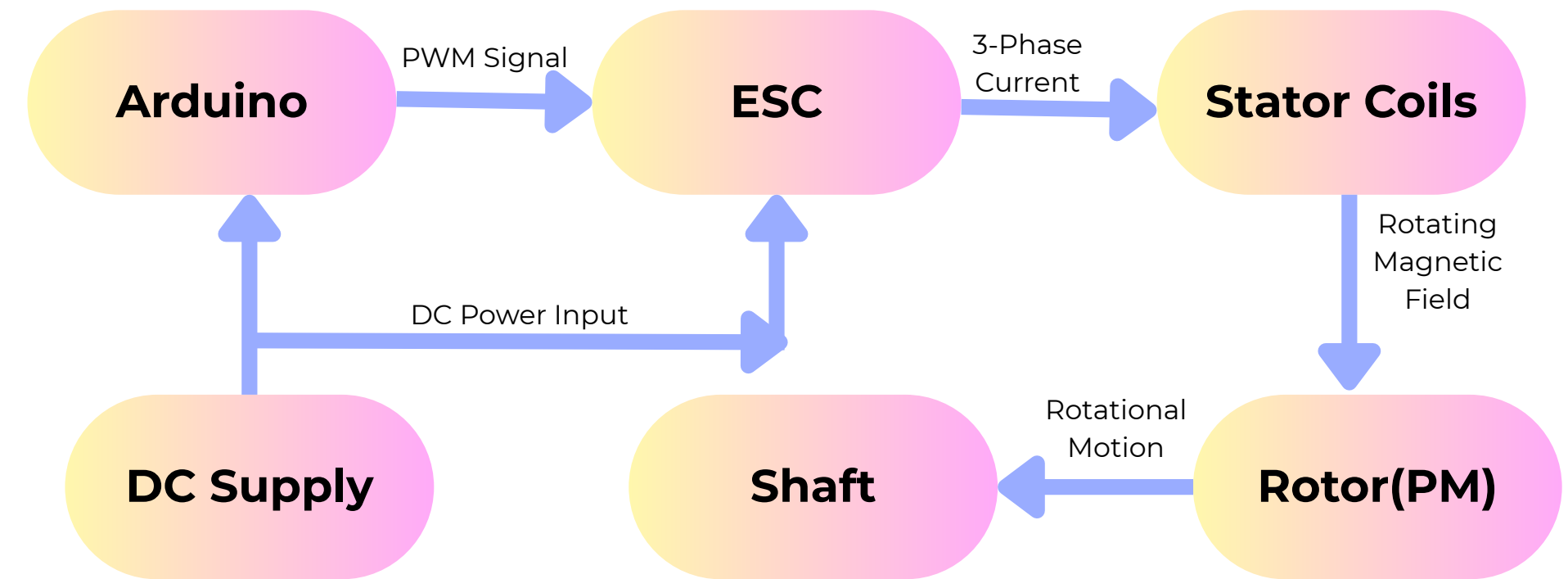
TEAM MEMBER	CONTRIBUTION
SUMIT KUMAR CHAUDHARY 23EE02011	Proposed the BLDC motor idea, created the CAD model, and supported component selection
PRANAV ADITYA 23EE01045	Led the hardware implementation, designed and assembled the rotor, integrated Arduino-ESC setup, and handled fault detection
DEVANSH GUPTA 23EE01011	Co-led the hardware development, designed and built the stator, managed full wiring and mechanical integration, and assisted in testing and troubleshooting
SWASTIK PANDA 23EE01069	Performed design calculations, troubleshooting, fault detection and presentation/documentation.

UNDER THE GUIDANCE OF
DR. ANKIT DALAL

BRUSHLESS DC MOTOR

WORKING PRINCIPLE

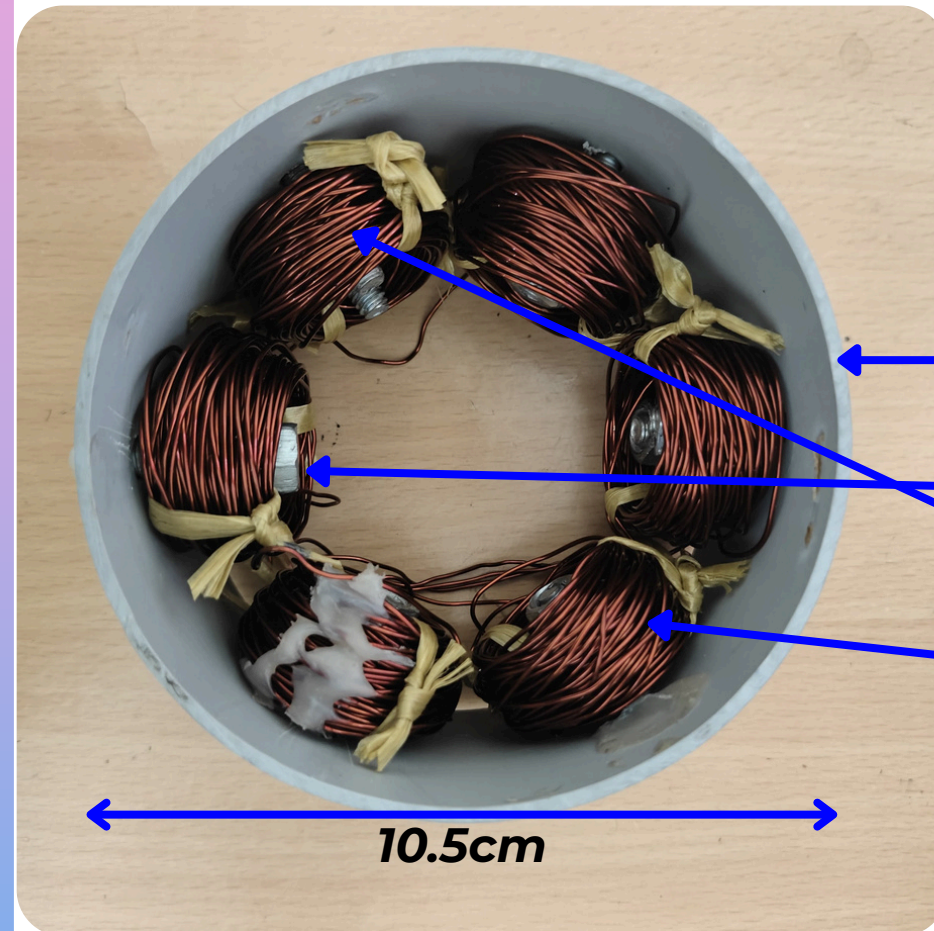
- Arduino sends PWM signals to the ESC.
- ESC energizes stator windings in sequence.
- A rotating magnetic field is created.
- Rotor (with magnets) follows the magnetic field.
- Rotation continues with synchronized switching.



ACTUAL MODEL & COMPONENTS USED

COMPONENTS	SPECIFICATION
Neodymium Magnets	8×1.5 mm, Circular
Copper Wire	22 SWG, Enamel Coated
Stator Core	PVC ring with 6 slots, iron bolts, star-connected copper coils
Rotor Material	Plastic disc with 4 poles, embedded neodymium magnets, wooden shaft
Shaft	Wooden stick, 17 cm length
Electronic Speed Controller (ESC)	30A Brushless ESC
Arduino Board	Standard Arduino UNO
Power Supply	Regulated DC Power Supply

DESIGNS AND CALCULATIONS



Stator Design

- 6-slot PVC ring
- Iron bolts as poles
- Star-connected double-layered windings
- 80 turns per coil, clockwise



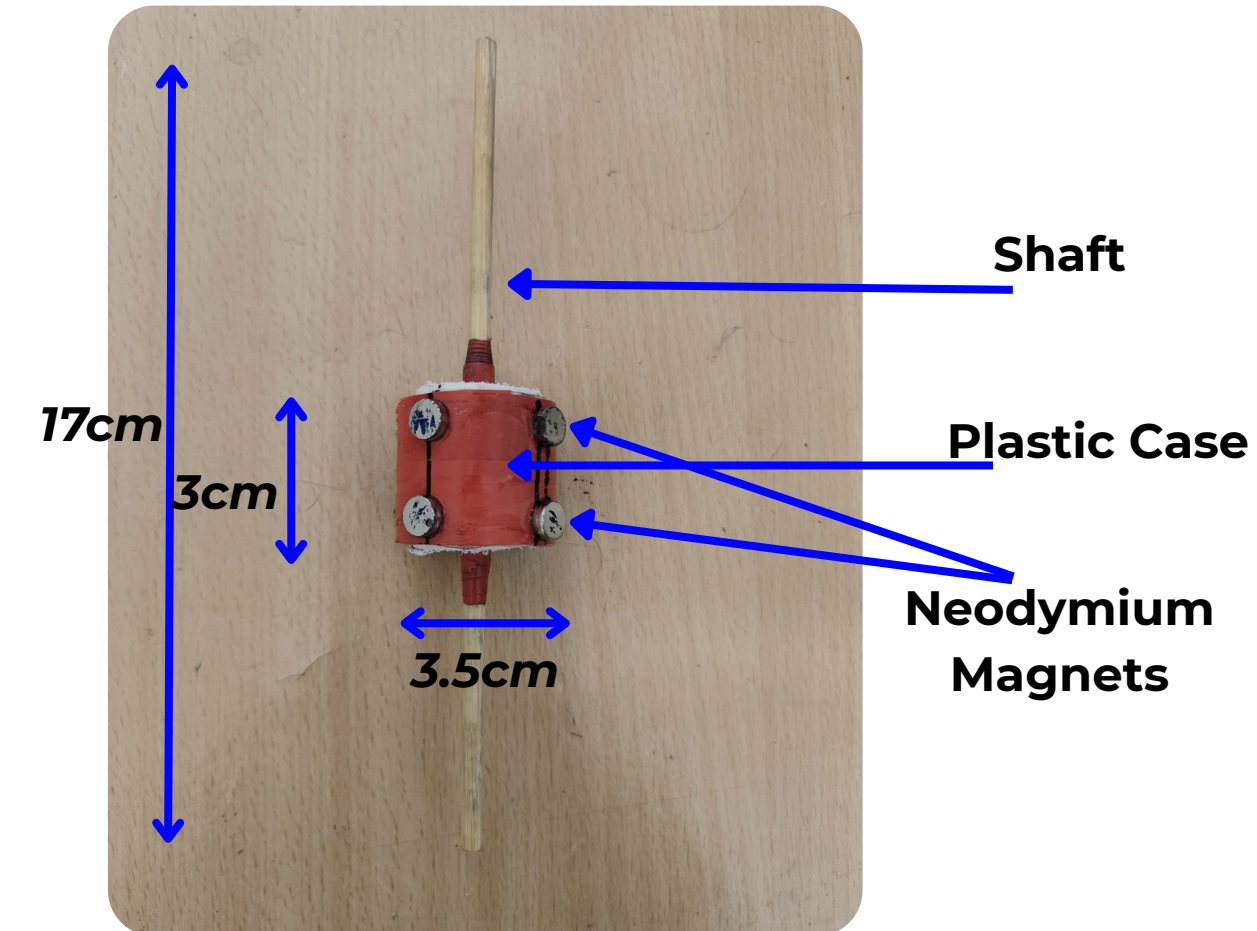
Winding Diagram

PVC Case

Iron Bolts As Teeth

Windings Of Same Phase

10.5cm



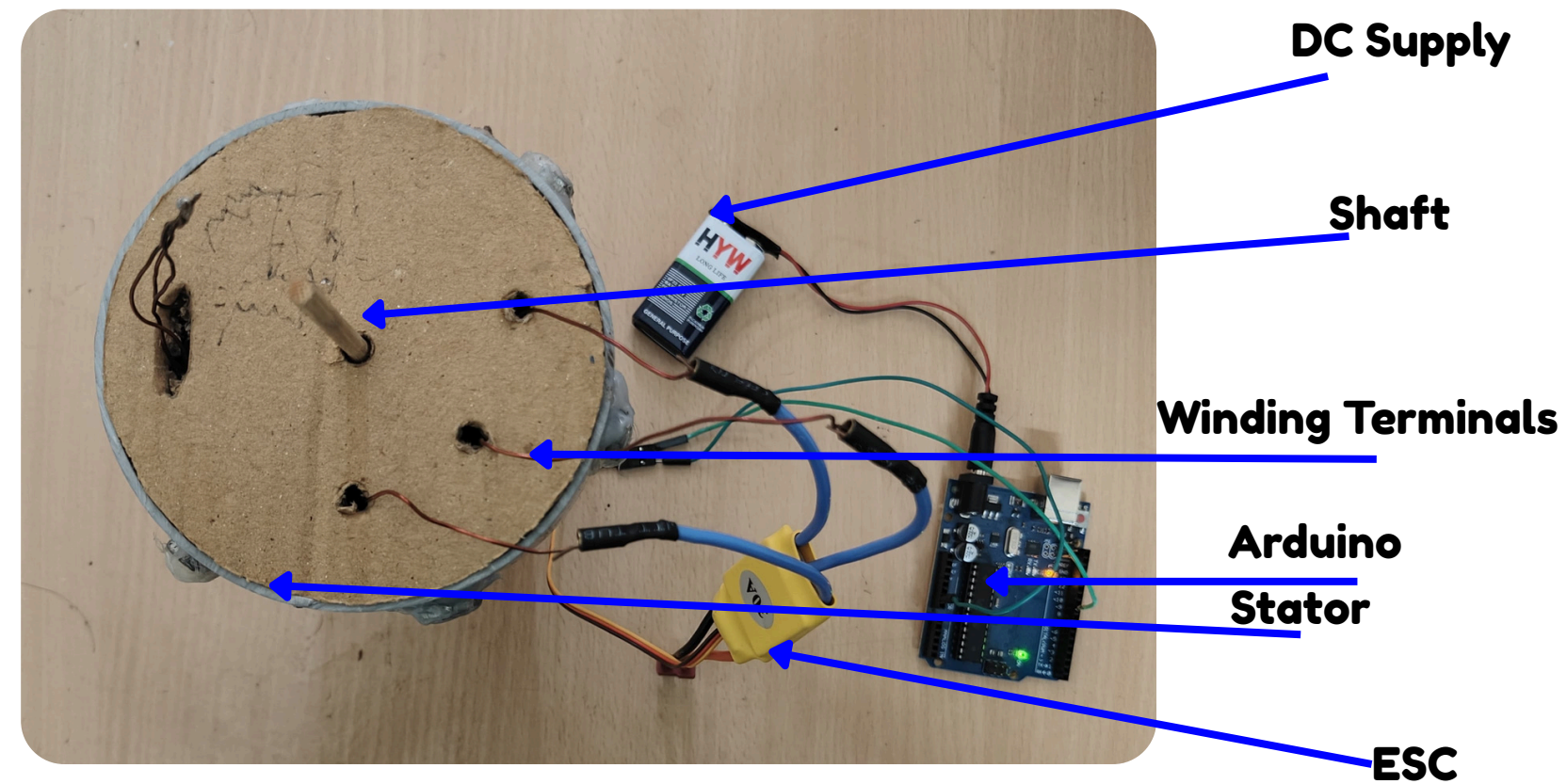
Rotor Design

- 4-pole plastic disc
- 8 neodymium magnets (8×1.5 mm)
- Wooden shaft at center

Key Parameters

- **Slots/Pole:** 6 slots / 4 poles
- **Winding:** Star-connected, 3-phase
- **Coils:** 6 (80 turns each, double layer)
- **Winding Factor:** ≈ 0.866
- **Phase Shift:** 120°
- **Observation:** Small sinusoidal voltages on DSO

DEMONSTRATIONS



Fully assembled BLDC motor with hand-made rotor, stator, and wiring.



3-phase sinusoidal back EMF observed during manual rotation.

- Final integration of rotor, stator, and windings
- Detected induced 3-phase sinusoidal EMF
- Minor vibration, no self-sustained rotation
- Verified phase shift and expected waveform

OBSERVATIONS, CHALLENGES AND FUTURE SCOPE

Observations

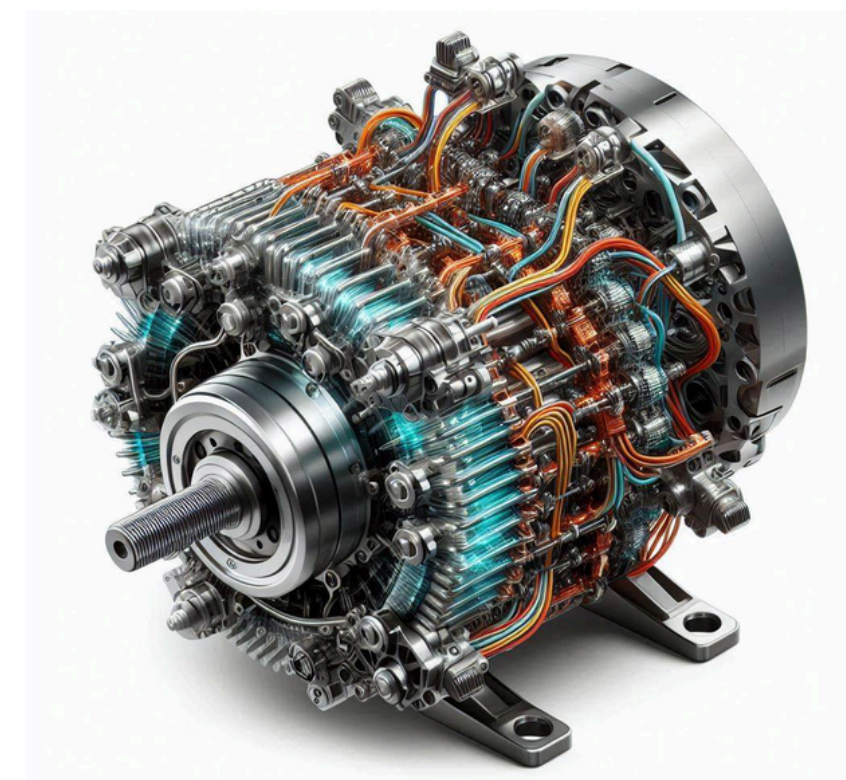
- ✓ Slight rotor vibration
- ✓ Induced 3-phase voltage
- ✓ Waveforms were ~sinusoidal and 120° phase-shifted.
- ✓ Accurate rotor-stator alignment and fitting

Challenges

- ⚠ Weak magnetic field
- ⚠ High friction at shaft-rotor interface.
- ⚠ Less precise coil placement
- ⚠ ESC voltage limitation

Future Scope

- ◆ Stronger magnets
- ◆ Hall sensors
- ◆ Better ball bearings
- ◆ 3D printed parts





THANK YOU

ANY QUESTIONS?

